Separating Training, Evaluation and Constraint-Set Noise in Fairness Uncertainty Quantification

Cenfa Lu
School of Mathematics and Computational Science,
Guilin University of Electronic Technology, Guilin 541004, China
joffin@163.com ORCID: 0009-0003-3481-3624
(corresponding author)

October 26, 2026

Abstract

Fairness-aware classifiers trained by stochastic gradient methods are random objects, so any reported demographic-parity or equalised-odds gap is random as well. Recent work constructs confidence intervals for such gaps with an online multiplier bootstrap, but those intervals quantify only the randomness of the optimisation trajectory. With the held-out sets used in practice, a second and often larger component—the sampling noise of the evaluation set—is ignored, and a third source of variability enters through the constraint set that defines the target of the estimate. We decompose the error of the reported gap into its training and evaluation parts, derive the limiting distribution and the exact coverage loss of the existing interval as a function of the ratio between training iterations and evaluation size, and prove that it is valid only when the evaluation set is large relative to the training run. We then propose a joint bootstrap that resamples the training trajectories and the evaluation set—and, when the audit design itself is random, adds a constraint-set layer—prove its asymptotic validity for a fixed audit design, and give a sample-size allocation rule. In a reproduction on synthetic data, Adult, ACSIncome and COMPAS, the existing interval covers the population fairness gap with probability between 0.00 and 0.62, whereas the joint interval attains nominal coverage (on Adult, 0.98 versus 0.30). A variance budget shows that training noise accounts for only 0.8%–8.3% of the variability of a reported gap, with evaluation and constraint-set sampling dominating.

Keywords: fairness auditing; uncertainty quantification; constrained stochastic optimization; online bootstrap; distribution-free inference.

1 Introduction

Machine-learning models used in high-stakes decisions are increasingly required to report fairness diagnostics such as the demographic-parity (DP) gap or the equalised-odds (EO) gap between protected groups [Hardt et al., 2016, Corbett-Davies et al., 2017]. In practice these diagnostics are computed from a finite held-out sample for a classifier whose parameters were themselves produced by a stochastic optimisation algorithm. Both steps introduce randomness, and a point estimate of a fairness gap without a measure of uncertainty is therefore of limited use for auditing.

Roy and Mohapatra [2026] take up this problem for fairness-aware linear classifiers trained online by stochastic dual averaging (for DP) or penalised SGD (for EO-type criteria). Viewing training as constrained stochastic optimisation, they establish asymptotic normality of the averaged iterate, and because the asymptotic covariance depends on the unknown optimum they propose an online

multiplier bootstrap operating on a single data stream. The resulting interval is appealing: it is computationally cheap, parallelisable, and requires no repeated training.

This paper starts from a simple observation about what such an interval actually covers. Write θ^* for the population optimum of the fairness-aware objective, $\bar{\theta}_T$ for the averaged online iterate after T steps, and $\hat{\Delta}_m$ for the empirical fairness gap computed on an independent held-out set $\tilde{D}$ of size m. The reported number is $\hat{\Delta}_m(\bar{\theta}_T)$, while the quantity a deployment decision depends on is the population gap $\Delta(\theta^*)$. The error splits into

$$\hat{\Delta}_m(\bar{\theta}_T) - \Delta(\theta^*) = \underbrace{\Delta(\bar{\theta}_T) - \Delta(\theta^*)}_{\text{training}} + \underbrace{\hat{\Delta}_m(\bar{\theta}_T) - \Delta(\bar{\theta}_T)}_{\text{evaluation}},$$

and the two terms are independent because $\tilde{D}$ is independent of the training stream. The online bootstrap approximates the law of the first term only. If the evaluation error is small relative to the training noise the distinction is immaterial; in the operating regime of interest it is not. In our reproduction, with $m = 1000$ held-out observations, the evaluation noise for Adult is 3.9 times the training noise, and the existing interval covers the population gap in 30% of repetitions instead of the nominal 95%.

Contributions.

1. *Decomposition and a diagnostic.* We show that the error of the reported gap relative to a fixed audit design decomposes into two asymptotically independent components—training and evaluation sampling—and that a third, design component arises when the constraint set itself is a random draw; we derive the limiting distribution (Theorem 1) together with the exact coverage of the existing interval (Corollary 1). The interval is asymptotically valid only when $m \gg T\,v^2/\sigma_\Delta^2$, a condition violated by orders of magnitude in published experimental setups.

2. *A joint bootstrap.* We propose to combine the multiplier bootstrap over training trajectories with a resampling of the evaluation set, and to add an outer bootstrap over the constraint set when the design itself is random; the two-source construction is asymptotically valid for a fixed audit design (Theorem 2), costs one extra pass over the evaluation set, and the design layer adds only a small number of direct constrained solves.

3. *A variance budget and design guidance.* On Adult, ACSIncome and COMPAS the training component accounts for only 0.8–8.3% of the variability of a reported gap; the dominant components are evaluation sampling (37–78%) and constraint-set sampling (18–55%). We translate this into a rule for allocating audit effort between training iterations, labelled evaluation data and the choice of constraint set.

4. *A reproducible study.* We re-implement the baseline from the published equations (no official code was released), reproduce its qualitative claims on five data settings, and document two further reproducibility findings: the published constant step size does not reach the reported convergence regime, and the synthetic data generator is close to degenerate. All code, data splits and raw results are released.

The remainder of the paper is organised as follows. Section 2 reviews related work, Section 3 fixes notation, Section 4 states the decomposition, Section 5 describes the joint bootstrap, Section 6 reports experiments and Section 7 discusses limitations and practical implications.

2 Related work

Online inference for stochastic approximation. Asymptotic normality of averaged stochastic approximation goes back to Polyak and Juditsky [1992]; constrained variants are analysed by Duchi and Ruan [2021]. Online bootstrap and batch-means procedures for unconstrained SGD are developed by Fang et al. [2018], Chen et al. [2020], Zhu et al. [2023] and extended to policy evaluation [Ramprasad et al., 2023]. These methods target the model parameter; for fairness we need the law of a non-smooth functional of it, which is why Roy and Mohapatra [2026] combine a functional delta-method argument with the multiplier bootstrap. Our work does not modify the training-side theory; it adds the evaluation component on top of it and treats the constraint-set (design) component separately, as variability of the target rather than of the estimate.

Statistical inference for fairness. A parallel literature tests whether a *fixed* classifier is fair: permutation tests [DiCiccio et al., 2020], optimal-transport projections [Taskesen et al., 2021], adversarial individual-fairness tests [Maity et al., 2021], and bootstrap audits over rich group collections [Cherian and Candès, 2024]. These papers treat the classifier as given and quantify evaluation uncertainty only, which is the complementary half of our decomposition. Our method targets the composition of training and evaluation uncertainty and makes the two components separately reportable.

Uncertainty in fairness more broadly. Rosenblatt and Witter [2024] benchmark uncertainty estimates used *inside* fair predictors, a different object from uncertainty about the fairness of a trained predictor. Moreau and Weber [2024] argue conceptually for uncertainty-aware fairness reporting. Subgroup-level fairness estimates also carry estimation variance that grows as the subgroup shrinks; our evaluation component formalises this in the DI/DM setting and is combined with the training component rather than studied in isolation.

Reproducibility and its use here. Neither the UAI proceedings version nor the earlier arXiv version [Roy and Mohapatra, 2023] ships official code, so we re-implemented both training procedures and the multiplier bootstrap from the published equations and report a full reproduction study. The qualitative claims reproduce: the mean interval score decreases with the number of iterations, the empirical distribution of the estimated gap converges to the benchmark normal law in Wasserstein-2 distance, and the interval covers the held-out estimand. Two discrepancies are documented: the published constant step size ($\eta_k = 0.01k^{-0.501}$) does not reach the convergence regime displayed in the paper within 10^6 iterations, and the synthetic generator is close to degenerate because the protected attribute is nearly a copy of the label. We therefore calibrate the step size (and, on real data, the regularisation strength) and report both the paper-literal and the calibrated settings.

3 Problem setup

Data and classifier. Let $(x, y, z) \sim P$ with features $x \in \mathbb{R}^p$, label $y \in \{-1, 1\}$ and protected attribute $z \in \{0, 1\}$. We consider linear classifiers $h(x; \theta) = \text{sign}(\theta^\top \varphi(x))$ with $\varphi(x) = (x^\top, 1)^\top \in \mathbb{R}^d$, $d = p + 1$.

Table 1: Notation.

symbol	meaning
θ	linear-classifier parameter; $h(x;\theta) = \mathrm{sign}(\theta^\top \varphi(x))$
θ^*	population constrained optimum (1)
$\bar{\theta}_T$	averaged online iterate after T steps
D', n_c	constraint set (audit design) and its size
$\tilde{D}$, m	held-out evaluation set and its size
$\Delta(\theta)$, $\hat{\Delta}_m(\theta)$	population and empirical fairness gap
σ^2_Δ, v^2	training and evaluation variance constants
$\lambda = T/m$	training-to-evaluation effort ratio
σ^2_{tot}	$\sigma^2_\Delta + \lambda v^2$, limiting variance of the reported gap
B	number of multiplier-bootstrap trajectories
η_k, κ, R	step size, L_2 regularisation, mistreatment penalty

Disparate impact. With tolerance $\epsilon > 0$ and a held-out constraint set D' of size n_c, define $\tilde{x} = n_c^{-1} \sum_{i \in D'} (z_i - \bar{z}) \varphi(x_i)$ and $\Theta(D') = \{\theta : |\tilde{x}^\top \theta| \le \epsilon\}$. The target parameter is

$$\theta^* = \arg \min_{\theta \in \Theta(D')} \ell(\theta), \qquad \ell(\theta) = \mathbb{E}[\log(1 + e^{-y\theta^\top \varphi(x)})] + \frac{\kappa}{2}\|\theta\|_2^2, \tag{1}$$

and the population gap is $\Delta(\theta) = P(h = 1 \mid z = 0) - P(h = 1 \mid z = 1)$ with $\phi(\theta) = |\Delta(\theta)|$.

Disparate mistreatment. With penalty $R^2 > 0$ and $D'_- = \{i \in D' : y_i = -1\}$, $\gamma(\theta; D'_-) = \left(n_-^{-1} \sum_{i \in D'_-} (z_i - \bar{z}) \min(0, y_i \varphi(x_i)^\top \theta)\right)^2$ and $\theta^* = \arg\min_\theta \ell(\theta) + R^2 \gamma(\theta; D'_-)$; the population gap conditions on $y = -1$.

Online training. The averaged iterate $\bar{\theta}_T = T^{-1} \sum_{k=1}^T \theta_k$ is produced by stochastic dual averaging with projection onto $\Theta(D')$ for DP, or by penalised SGD for EO. An online multiplier bootstrap with B perturbed trajectories, each driven by the same single data stream with i.i.d. multipliers $V \sim \mathrm{Unif}[1 - \sqrt{3}, 1 + \sqrt{3}]$, yields $\{\hat{\theta}^b_T\}_{b=1}^B$ and hence $\{\hat{\Delta}_m(\bar{\theta}^b_T)\}$; see Roy and Mohapatra [2026] for the algorithm and its training-side guarantees.

Two estimands. Notation is collected in Table 1. We distinguish the *held-out estimand* $\hat{\Delta}_m(\theta^*)$, i.e. the fairness of the optimal classifier evaluated on the very sample used by the auditor, from the *population estimand* $\Delta(\theta^*)$, i.e. the fairness of the deployed classifier on the data distribution. The distinction is vacuous only when $m \to \infty$; a confidence interval must name its target.

4 Decomposition: two error sources and a design component

Assumption 1 (inherited from Roy and Mohapatra, 2026)**.** *The loss is strongly convex and smooth on a neighbourhood of θ^*; the data stream is i.i.d.; the decision boundary is linear in θ; the conditional feature distributions admit continuous densities at the boundary. The evaluation set $\tilde{D}$ is drawn i.i.d. from P independently of the training stream and of D'.*

All statements in this section condition on the realised constraint set D'. Randomness in D' enters through the target θ^* rather than through the estimator, and is analysed separately in the variance budget of Section 6.

Lemma 1 (training noise; Roy and Mohapatra, 2026). *Under Assumption 1,* $\sqrt{T}(\Delta(\bar{\theta}_T)-\Delta(\theta^*)) \Rightarrow \mathcal{N}(0,\sigma_\Delta^2)$ *with* $\sigma_\Delta^2 = g^\top \Sigma g$, $g = \nabla\Delta(\theta^*)$, *and* Σ *the asymptotic covariance of the averaged iterate.*

Lemma 2 (evaluation noise). *Under Assumption 1, uniformly over* θ *in a neighbourhood of* θ^*,

$$\sqrt{m}\big(\hat{\Delta}_m(\theta) - \Delta(\theta)\big) \Rightarrow \mathcal{N}\big(0,\, v^2(\theta)\big), \qquad v^2(\theta) = \frac{p_0(1-p_0)}{P(z=0)} + \frac{p_1(1-p_1)}{P(z=1)},$$

with $p_j = P(h=1 \mid z=j)$, *and the analogous expression with* $y=-1$ *for DM. Moreover the two convergences are independent.*

Theorem 1 (limit distribution). *Suppose* $\Delta(\theta^*) \neq 0$ *and* $T/m \to \lambda \in [0,\infty)$. *Then under Assumption 1,*

$$\sqrt{T}\big(\hat{\Delta}_m(\bar{\theta}_T) - \Delta(\theta^*)\big) \Rightarrow \mathcal{N}\big(0,\ \sigma_\Delta^2 + \lambda v^2\big), \qquad \textit{conditionally on } D'.$$

Corollary 1 (coverage of the existing interval). *The fixed-time interval of Roy and Mohapatra [2026], of asymptotic half-width* $z_{\alpha/2}\sigma_\Delta/\sqrt{T}$, *has asymptotic coverage*

$$2\Phi\Big(z_{\alpha/2}\,\frac{\sigma_\Delta}{\sqrt{\sigma_\Delta^2 + \lambda v^2}}\Big) - 1,$$

which equals $1-\alpha$ *if and only if* $\lambda v^2/\sigma_\Delta^2 \to 0$, *i.e.* $m \gg Tv^2/\sigma_\Delta^2$.

Remark 1 (two coverage protocols). *If the evaluation set* $\tilde{D}$ *is held fixed across repetitions, as in the original simulation design, the evaluation error becomes a constant bias* $b = \hat{\Delta}_m(\theta^*) - \Delta(\theta^*)$ *and coverage is* $\Phi(z_{\alpha/2} - |b|\sqrt{T}/\sigma_\Delta) - \Phi(-z_{\alpha/2} - |b|\sqrt{T}/\sigma_\Delta)$. *Coverage statements about population fairness must therefore be evaluated with a fresh evaluation set in each repetition; we report both protocols.*

5 Method: joint bootstrap

For $b = 1,\dots,B$ multiplier trajectories, $r = 1,\dots,R$ evaluation resamples, and $s = 1,\dots,S$ constraint-set resamples, define the bootstrap deviations

$$D^*_{b,r,s} = \underbrace{\hat{\Delta}_m(\bar{\theta}^b_T) - \hat{\Delta}_m(\bar{\theta}_T)}_{\text{training}} + \underbrace{\hat{\Delta}_{m,r}(\bar{\theta}_T) - \hat{\Delta}_m(\bar{\theta}_T)}_{\text{evaluation}} + \underbrace{\hat{\Delta}_m(\theta^*_s) - \hat{\Delta}_m(\theta^*)}_{\text{constraint set}},$$

where the third term uses the constrained optimum θ^*_s computed for a resampled constraint set D'_s; it is obtained by solving the constrained problem directly, with no retraining. The proposed interval is $\big[\hat{\Delta}_m(\bar{\theta}_T) + \hat{q}_{\alpha/2},\ \hat{\Delta}_m(\bar{\theta}_T) + \hat{q}_{1-\alpha/2}\big]$ for the quantiles of D^*, mapped to $|\Delta|$ by folding: if the resulting interval for the signed gap excludes zero we report the absolute values of its endpoints, otherwise we report $[0, \max(|l|,|u|)]$. When the constraint set is regarded as fixed by design, the third term is omitted and the interval targets $\Delta(\theta^*)$ for the realised D'. All experiments below use this fixed-design version, so the coverage validated there refers to the two inference-error components; the design layer is quantified separately in the variance budget.

Theorem 2 (asymptotic validity). *Under Assumption 1, if* $\bar{\theta}_T \to \theta^*$ *and the constraint set is fixed by design (third term omitted), then conditionally on the data the law of* D^* *converges in probability to the law of* $\hat{\Delta}_m(\bar{\theta}_T) - \Delta(\theta^*)$; *consequently the percentile interval has asymptotic coverage* $1-\alpha$ *for* $\phi(\theta^*)$. *If the design is random and the third term is included, the conclusion holds for the design-conditional target and the interval is conservative for a realised design, because the added term has mean zero and is asymptotically independent of the first two.*

Cost. The evaluation resampling is a vectorised operation on the m held-out predictions and costs $O(Rm)$. The training deviation reuses the B trajectories already produced by the online bootstrap. The constraint-set bootstrap requires S direct constrained solves, each costing a few iterations of a quasi-Newton method, and no additional passes over the data stream.

Sample-size allocation. With per-unit costs c_T and c_m and target half-width w, minimising $c_T T + c_m m$ subject to $z_{\alpha/2}^2(\sigma_\Delta^2/T + v^2/m) \le w^2$ gives

$$\frac{m}{T} = \sqrt{\frac{v^2 c_T}{\sigma_\Delta^2 c_m}}, \qquad T = \frac{z_{\alpha/2}^2}{w^2}\left(\sigma_\Delta^2 + \sqrt{\sigma_\Delta^2 v^2 c_T / c_m}\right).$$

Because σ_Δ^2 and v^2 are of the same order in the regimes we study, the binding constraint is the *absolute* size of m: increasing T alone cannot push the half-width below $z_{\alpha/2} v/\sqrt{m}$ (Figure 3).

6 Experiments

Setup and protocols. Synthetic data follow the generators of Zafar et al. [2019] used by Roy and Mohapatra [2026]; real data are Adult and ACSIncome [Ding et al., 2021] and COMPAS, with the sensitive attributes and tolerances of the original Table 1. Three protocols appear below. *(R) Reproduction*: one fixed constraint set and one fixed held-out set, repetitions resample only the training stream, which is the protocol of the original simulation study. *(A) Protocol A*: the constraint set is fixed but every repetition draws a fresh evaluation set from a disjoint reservoir, so that coverage is evaluated over the evaluation sampling law as well. *(B) Budget*: a single run with the reported audit budget ($T = 3 \times 10^4$, $m = 1000$ for Adult/ACSIncome and $m = 400$ for COMPAS), used for the sample-size analysis. Unless stated otherwise $B = 100$ multiplier trajectories and $\alpha = 0.05$; step size and regularisation are calibrated as described above ($\eta_0 = 0.1$ for DI, 0.03 for the synthetic DM generator, $\kappa = 0.1$). The constraint set is held fixed at the sampled audit design in all three protocols, so the joint interval used below is the two-source version; the constraint-set layer is quantified separately in the variance budget.

Reproduction of the baseline. Table 2 summarises the reproduced behaviour in the original fixed-held-out protocol. The qualitative claims of the baseline hold: the mean interval score decreases with T, the empirical distribution converges to the benchmark normal law, and the interval covers the held-out estimand. For the population estimand, however, coverage ranges from 0.00 to 0.62 except in the synthetic DM setting, where the interval is wide enough by accident. Figures 1 and 2 show the per-iteration trajectories behind these numbers: convergence to the asymptotic law is fast on the synthetic experiment, the interval covers the held-out estimand throughout, and the population estimand is missed as soon as the evaluation error is comparable to the interval width. The $T^{-1/2}$ rate behind Theorem 1 is itself visible in a separate grid of training horizons: the distance of $\bar{\theta}_T$ to the constrained optimum decays with log–log slope 0.47 and the signed training bias of the gap with slope 0.46 (see the empirical convergence rates in the supplementary material).

Protocol-A coverage. Table 3 varies the held-out size m with a fresh evaluation set per repetition. The baseline coverage increases with m exactly as predicted by Corollary 1, while the joint bootstrap stays at or above nominal coverage; its width converges to that of the baseline as m grows, quantifying the price of valid inference. The same pattern holds on the three real datasets, where the baseline interval attains 0.25–0.62 coverage and the joint interval 0.98–1.00.

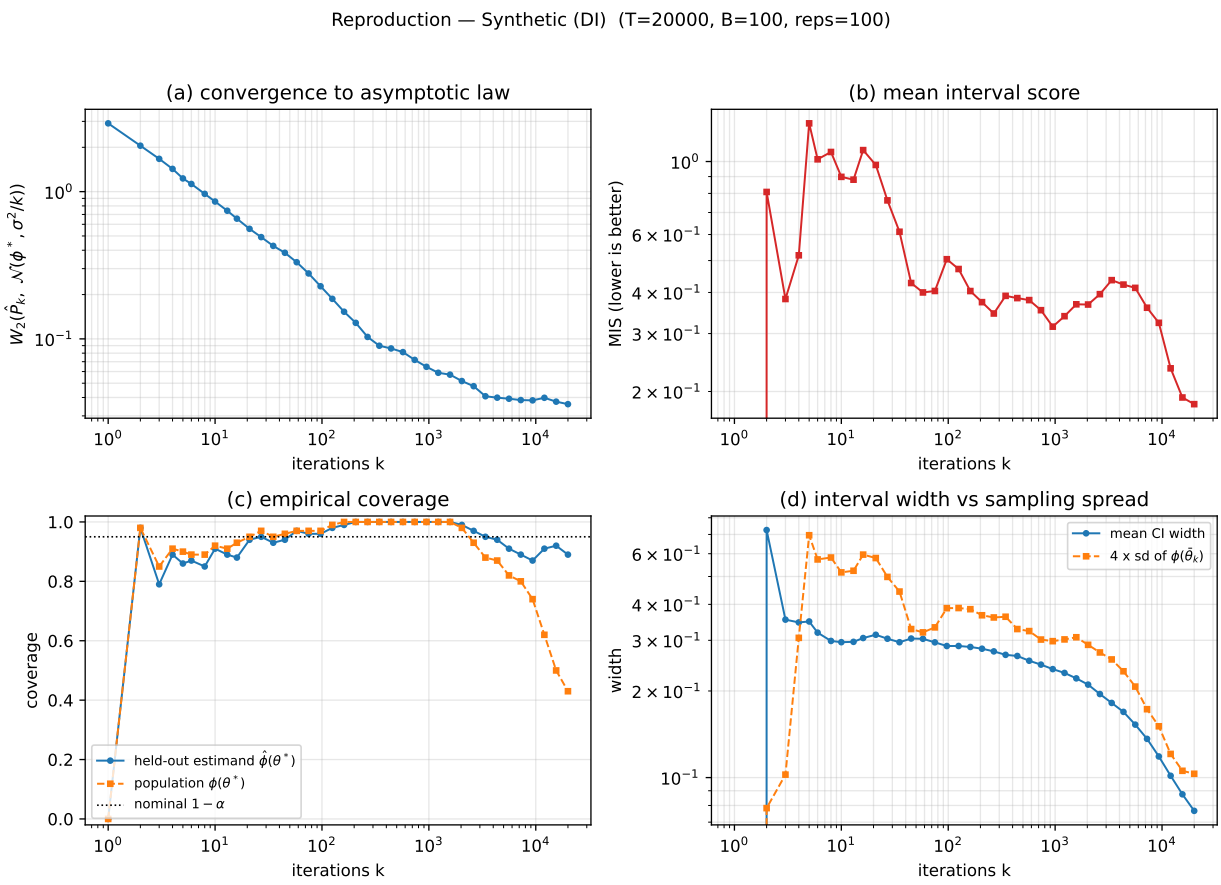


Figure 1: Reproduction of the synthetic DI experiment: (a) Wasserstein distance between the empirical law of $\hat{\Delta}_m(\bar{\theta}_T)$ and its asymptotic limit; (b) mean interval score; (c) coverage of the held-out estimand $\hat{\Delta}_m(\theta^*)$ and of the population estimand $\Delta(\theta^*)$; (d) interval width against the sampling spread $4v/\sqrt{m}$. The interval covers the held-out estimand but not the population estimand, the failure mechanism analysed in Section 4.

Variance budget. Table 4 separates the three sources of variability in a reported gap. The training and evaluation components are errors of the reported number around a fixed target; the constraint-set component measures how far the target itself moves when the audit design is redrawn, and is therefore a design sensitivity rather than an inferential uncertainty. Training noise, the only component addressed by the baseline, contributes between 0.8% and 8.3% of the total. The design component is particularly severe for COMPAS, where resampling D' changes the sign of the reported gap, so a single constrained run is not a reliable audit without a design-sensitivity statement.

Sample-size planning and the minimum certifiable gap. Table 4 and the planning formula of Section 5 together answer a practical question: at a given audit budget, which fairness gaps can be certified at all? Using the component estimates from the protocol-A runs, the best achievable 95% half-width at $T = 3 \times 10^4$ and $m = 1000$ is 0.053 for Adult, 0.060 for ACSIncome and 0.133 for COMPAS ($m = 400$). The reported population gaps are 0.059, 0.041 and 0.006 respectively, so at these budgets the audit can exclude a zero gap for Adult only; for ACSIncome and COMPAS the minimum certifiable gap exceeds the gap itself, by a factor of 1.5 and 21, respectively. Figure 3

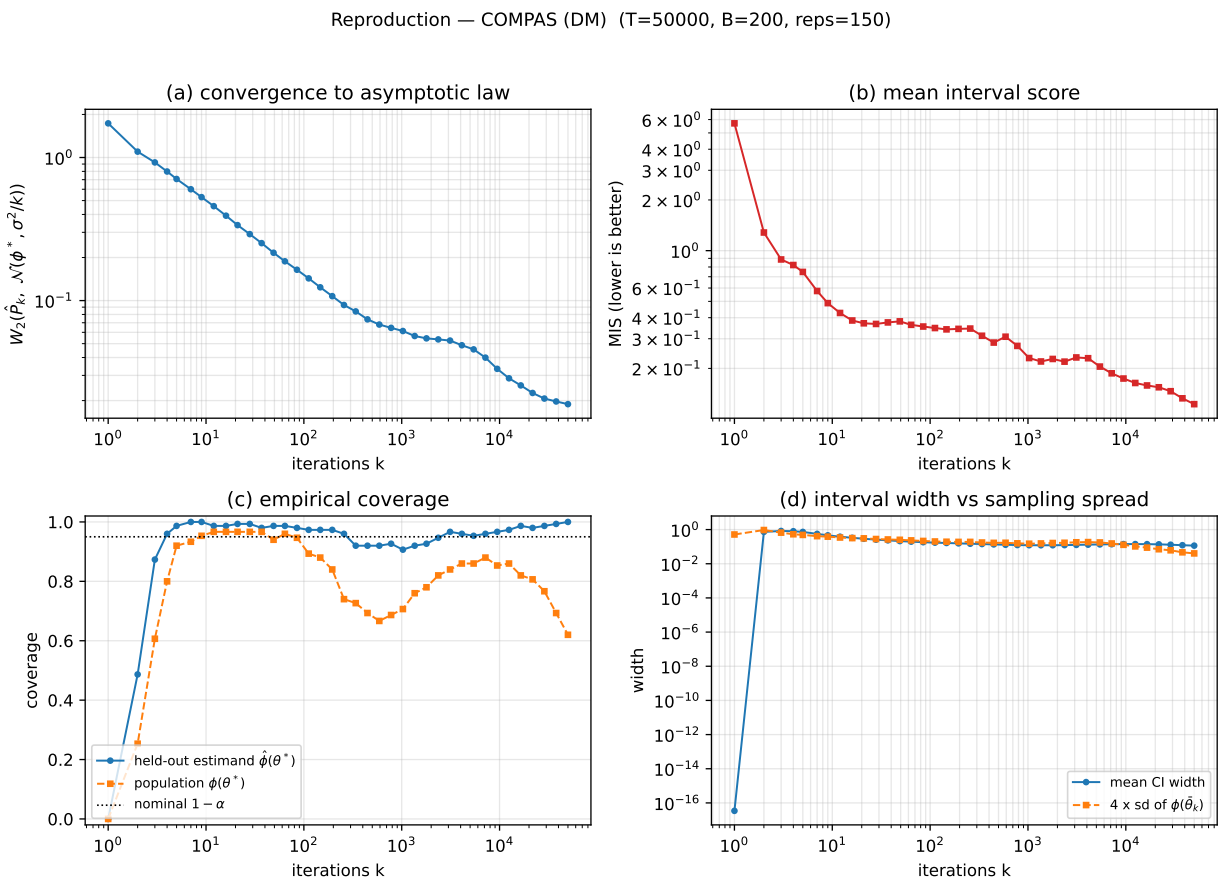


Figure 2: Reproduction of the COMPAS DM experiment (same four panels). The penalty term is frozen at its training value inside the feasible region, so the averaged iterate stops short of the constrained boundary; the held-out estimand is covered because the evaluation term dominates, while the population estimand is covered in only 0.62 of repetitions.

shows the feasible (T, m) region for several target half-widths. The practical implication is that increasing T alone is not enough: because σ_Δ^2 and v^2 are of the same order, the half-width floor $z_{\alpha/2}v/\sqrt{m}$ sets a hard limit that only more evaluation data can lower.

Optimisation bias. A variance-based interval presumes that the averaged iterate has reached the neighbourhood of θ^*. Table 5 varies the regularisation strength on Adult under protocol (R). At $\kappa = 10^{-4}$ the optimum has norm 10.9 and the averaged iterate is 10.5 away from it: no interval based on the training variance can be valid. The table also shows that coverage is governed by the *functional* bias of ϕ, not by the parameter distance: at $\kappa = 10^{-2}$ the parameter bias is 0.81 yet coverage is nominal, while at $\kappa = 0.1$ a parameter bias of 0.12 is enough to lose all coverage because the DI functional shifts by more than the half-width (DP/DM gaps are discontinuous in θ). Audits should therefore report a convergence diagnostic together with the interval, and the optimisation bias must be controlled by the design (T and κ), not by the bootstrap.

Which source must be resampled? Table 6 isolates the two resampling components under protocol (A). The baseline resamples only training trajectories, an "evaluation-only" interval re-

Table 2: Reproduction of the baseline in the fixed-held-out protocol. "coverage held-out" is the fraction of repetitions whose interval contains $\hat{\Delta}_m(\theta^*)$; "coverage population" uses $\Delta(\theta^*)$ evaluated on the full data.

setting	T	reps	$\phi(\theta^*)$	cov. held-out	cov. population	width
Synthetic (DI)	2e4	100	0.0835	0.89	0.43	0.077
Synthetic (DM)	2e4	100	0.1401	0.86	1.00	0.208
Adult (DI)	5e4	100	0.0564	1.00	0.00	0.022
ACSIncome (DI)	5e4	150	0.0442	1.00	0.00	0.013
COMPAS (DM)	5e4	150	0.0133	1.00	0.62	0.115

Table 3: Protocol-A coverage of the population gap (nominal 0.95) with a fresh evaluation set in every repetition and a fixed constraint set (two-source joint interval). Synthetic DI: $T = 2 \times 10^4$, $B = 100$, 100 repetitions, $\phi(\theta^*) = 0.0212$. Real data: $T = 3 \times 10^4$, $B = 100$, 60 repetitions.

setting	m	cov. base	cov. joint	w base	w joint	ratio
Synthetic (DI)	100	0.56	0.99	0.107	0.283	2.64
Synthetic (DI)	1000	0.85	0.98	0.077	0.116	1.50
Synthetic (DI)	5000	0.99	1.00	0.074	0.085	1.15
Adult (DI)	1000	0.30	0.98	0.025	0.102	4.15
ACSIncome (DI)	1000	0.25	1.00	0.015	0.082	5.72
COMPAS (DM)	400	0.62	1.00	0.096	0.173	2.08

samples the held-out predictions of the trained model, and the joint interval resamples both. On the synthetic generators, where the two components are of comparable size, neither single-source interval is valid at $m = 100$ (demographic parity: 0.56 and 0.98; disparate mistreatment: 0.57 and 0.81), and only the joint interval comes close to nominal (1.00 and 0.88). At $m = 1000$ the evaluation noise shrinks and the training-only interval remains short (0.85 for demographic parity, 0.78 for disparate mistreatment), whereas the joint interval reaches nominal level on both generators (0.98 and 0.99). On the real audits the evaluation component dominates, so the training-only interval collapses to 0.30 (Adult) and 0.25 (ACSIncome) — and to 0.62 on COMPAS, where the training component carries a larger share — while the evaluation-only interval is close to nominal (0.97, 1.00 and 0.97). The component that is resampled determines coverage and the component that is ignored determines failure. The evaluation-only interval is already almost as wide as the joint one (0.081 versus 0.082 on ACSIncome, 0.098 versus 0.102 on Adult, 0.154 versus 0.173 on COMPAS), so valid inference costs little beyond resampling the evaluation set; but only resampling both is at or near nominal coverage in every setting. Figure 4 shows the same comparison graphically: each single-source interval fails precisely where the component it ignores is large. A finite-sample grid over training horizons and evaluation sizes (Table S9 of the supplement) shows the same ordering away from the asymptotic regime: the baseline interval can fall well below nominal when the evaluation set is small relative to the training run, while the joint interval never falls below nominal in the same grid.

7 Discussion

Our results do not contradict the training-side theory of Roy and Mohapatra [2026]; they delimit its scope. Three practical recommendations follow. First, auditors should report which estimand an interval targets, and should not label an interval for the held-out gap as an interval for the deployed

Table 4: Variance budget: standard errors of the two error components (training, evaluation) and of the constraint-set (design) component that shifts the target, with shares of the total variability; the design column is a sensitivity of the audit, not an inferential uncertainty.

dataset	se(train)	se(eval)	se(D')	share(train)	share(eval)	share(D')
Adult (DI)	0.0056	0.0259	0.0124	3.6%	78.3%	18.1%
ACSIncome (DI)	0.0034	0.0304	0.0245	0.8%	60.3%	39.0%
COMPAS (DM)	0.0293	0.0621	0.0753	8.3%	37.1%	54.6%

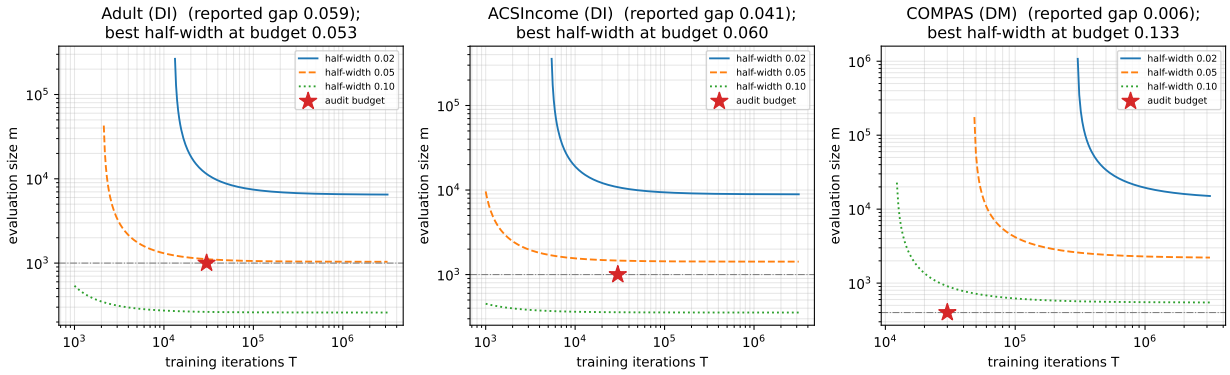


Figure 3: Feasible (T, m) region for target 95% half-widths (curves), with the audit budget ($\star$) and the reported population gap annotated in each panel. Points below the curve for a target width cannot certify that width.

classifier's fairness. Second, evaluation data should be budgeted before training compute: in our experiments $m = 1000$ labelled observations cannot certify a gap to better than a few percentage points regardless of T. Third, because the constraint set is itself a random audit design choice, a single constrained run should be accompanied by a constraint-set sensitivity analysis.

Limitations. The theory inherits the asymptotic regime, strong convexity and linear/NTK classifier assumptions of the baseline. The joint bootstrap requires $\bar{\theta}_T$ to be close to θ^*; unlike the evaluation and constraint-set components, the optimisation bias is *not* estimable from the same single-stream bootstrap, and we only report its diagnostic (distance to a directly computed optimum) and its effect (Table 5). The constraint-set component assumes that resampling D' is meaningful, that is, that the audit tolerances and group definitions are exchangeable across audits; when D' is a fixed policy object rather than a random audit sample, this component should be reported as a sensitivity analysis rather than folded into the interval. Finally, the synthetic generator supplied by the baseline is close to degenerate (the protected attribute is almost a copy of the label), so synthetic results should be read as a mechanism check rather than as evidence about realistic fairness–accuracy trade-offs. Extensions to multi-class criteria, intersectional group collections and neural classifiers are left for future work.

8 Conclusion

Reported fairness gaps of online-trained classifiers carry three kinds of noise. We showed that the existing online bootstrap interval quantifies only the training component, derived the exact rate at

Table 5: Optimisation-bias sensitivity on Adult (protocol R, 20 repetitions): the averaged iterate's distance to the constrained optimum, the population coverage of the baseline interval, and its width.

κ	$\|\theta^*\|$	median $\|\bar{\theta}_T - \theta^*\|$	coverage	CI width	$\phi(\theta^*)$
10^{-4}	10.89	10.50	0.00	0.0399	0.1394
10^{-2}	2.20	0.81	0.95	0.0382	0.0884
0.1	1.10	0.12	0.00	0.0289	0.0570
1.0	0.32	0.02	0.60	0.0314	0.0651

Table 6: Source ablation under protocol (A), nominal coverage 0.95: intervals that resample only training trajectories, only the evaluation set, or both. Synthetic generators: $T = 2 \times 10^4$, $B = 100$, 100 repetitions; real data: $T = 3 \times 10^4$, $B = 100$, 60 repetitions.

setting	m	cov. base	cov. eval	cov. joint	w base	w eval	w joint
Synthetic (DI)	100	0.56	0.98	1.00	0.107	0.270	0.283
Synthetic (DI)	1000	0.85	0.93	0.98	0.077	0.094	0.116
Synthetic (DM)	100	0.57	0.81	0.88	0.151	0.215	0.239
Synthetic (DM)	1000	0.78	0.71	0.99	0.139	0.073	0.158
Adult (DI)	1000	0.30	0.97	0.98	0.025	0.098	0.102
ACSIncome (DI)	1000	0.25	1.00	1.00	0.015	0.081	0.082
COMPAS (DM)	400	0.62	0.97	1.00	0.096	0.154	0.173

which its coverage degrades as a function of the ratio between training and evaluation effort, and proposed a joint bootstrap that restores nominal coverage by resampling the training trajectories and the evaluation set, with a constraint-set layer for audits whose design is itself random. A variance budget on three real audits shows that training noise accounts for less than a tenth of the variability of a reported gap in typical operating regimes: evaluation sampling dominates when the held-out set is small, and constraint-set sampling can be the largest component of all. The practical message is that fairness audits should budget evaluation data and audit-design sensitivity before training compute, and should report which estimand their interval targets. Future work includes neural classifiers, intersectional group collections and sequential audits.

Statements

Data availability. All datasets are public benchmarks (Adult, ACSIncome and COMPAS); the processing scripts, data splits, training and bootstrap code, and raw per-repetition result files are released in an open-source repository (MIT licence) that will be made publicly available on GitHub upon acceptance; a permanent repository DOI will be added at that stage. The reproduction report is included as Appendix B.

Code availability. The complete code base (data preparation, estimators, online and joint bootstrap, all experiment drivers, figure and table generation, and verification scripts) is available in the same repository under an MIT licence.

Funding. This research received no specific grant from any funding agency.

Competing interests. The author declares no competing interests.

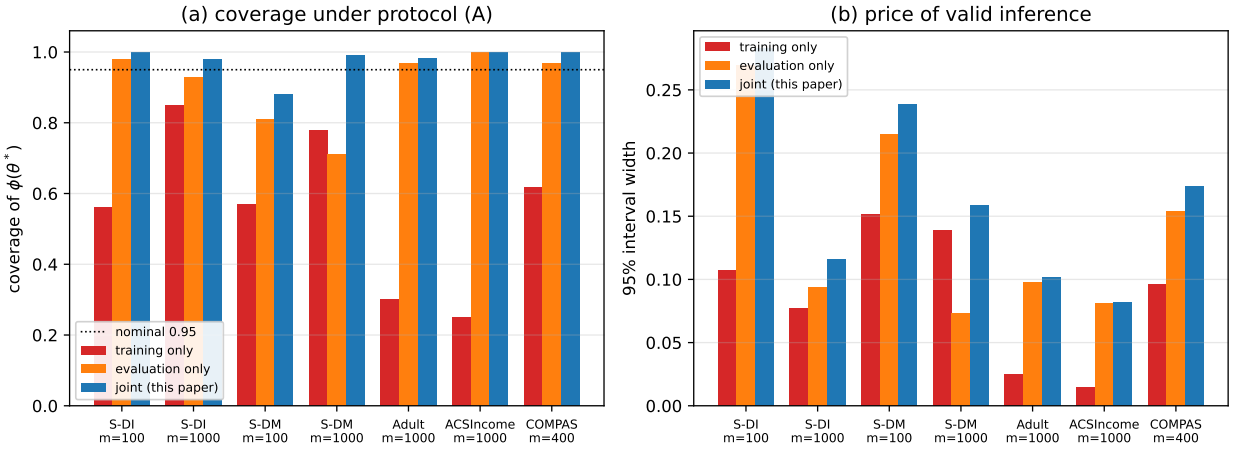


Figure 4: Source ablation under protocol (A) with nominal coverage 0.95: intervals that resample only the training trajectories, only the evaluation set, or both. (a) Coverage of the population gap; only the joint interval is at or near nominal level in every setting, while each single-source interval fails in the regime where the source it ignores dominates (training for the synthetic generators at $m = 100$, evaluation for the real audits). (b) The corresponding interval widths: valid inference costs little beyond resampling the evaluation set, since the evaluation-only and joint intervals nearly coincide on Adult and ACSIncome.

Ethics approval. Not applicable. The study uses publicly available secondary datasets (Adult, ACSIncome, COMPAS); no new human-subjects data were collected.

Consent to participate. Not applicable.

Consent for publication. Not applicable.

Authors' contributions. C.L. conceived the study, developed the theory, implemented the code, ran all experiments, and wrote the manuscript.

Acknowledgements. The author thanks the maintainers of the public data sets used in this study.

Use of AI tools. An AI-based language tool was used only to polish the English expression of the manuscript. It was not used to generate scientific content, code, or results, and the author verified every statement and number in the paper.

A Proofs

Throughout, $z_j = \mathbb{1}[z = j]$, $P_j = \Pr(z = j)$, $p_j = \Pr(h = 1 \mid z = j)$ and $\hat{P}_j = m^{-1} \sum_{i \in \tilde{D}} z_{i,j}$, $\bar{\varphi}_j = (\sum_{i \in \tilde{D}} z_{i,j} \varphi_i) / \sum_{i \in \tilde{D}} z_{i,j}$.

A.1 Proof of Lemma 2

Write $A = \sum_i z_{i,1}\varphi_i$, $B = \sum_i z_{i,1}$, $C = \sum_i z_{i,0}\varphi_i$, $D = \sum_i z_{i,0}$. Then $\hat{\Delta}_m(\theta) = A/B - C/D$, $A/m \to P_1 p_1$ and $B/m \to P_1$. Direct computation gives $\mathrm{Var}(A/m) = P_1 p_1(1 - P_1 p_1)$, $\mathrm{Var}(B/m) = P_1(1 - P_1)$ and $\mathrm{Cov}(A/m, B/m) = P_1 p_1(1 - P_1)$, and analogously for (C, D). The delta method with $J = (1/P_1, -p_1/P_1^2, -1/P_0, p_0/P_0^2)$ yields

$$v^2(\theta) = \frac{p_1(1-p_1)}{P_1} + \frac{p_0(1-p_0)}{P_0},$$

which also follows from the law of total variance by conditioning on the realised group memberships. Independence of $\tilde{D}$ from the training stream and D' is part of Assumption 1. For uniformity, the class of half-space indicators is a bounded VC class; on a fixed neighbourhood of θ^* it is Donsker, so the empirical process is $O_p(m^{-1/2})$ uniformly in θ. On shrinking neighbourhoods, $P(\varphi_\theta \neq \varphi_{\theta^*}) \leq L\|\theta - \theta^*\|$ for some $L < \infty$ by the continuity of the feature densities at the boundary, and the same symmetrisation argument (van der Vaart and Wellner, 1996, Ch. 2.5) gives a local oscillation of order $o_p(m^{-1/2}) + \|\theta - \theta^*\|$ uniformly over $\|\theta - \theta^*\| \leq \delta_m$ for any $\delta_m \to 0$; this is what the linearisation step below uses. The DM case only replaces the indicator $h = 1$ by $h = -1$ and yields a finite sum over the two label strata.

A.2 Proof of Theorem 1

Add and subtract the evaluation estimand and split the error into three terms,

$$\sqrt{T}\big(\hat{\Delta}_m(\bar{\theta}_T) - \Delta(\theta^*)\big) = \underbrace{\sqrt{T}\big(\Delta(\bar{\theta}_T) - \Delta(\theta^*)\big)}_{(i)} + \underbrace{\sqrt{T}\big(\hat{\Delta}_m(\theta^*) - \Delta(\theta^*)\big)}_{(ii)} + \underbrace{\sqrt{T}\big[(\hat{\Delta}_m - \Delta)(\bar{\theta}_T) - (\hat{\Delta}_m - \Delta)(\theta^*)\big]}_{(iii)}.$$

Term (i) converges to $\mathcal{N}(0, \sigma_\Delta^2)$ by Lemma 1. For (iii), a first-order Taylor expansion of the empirical process at θ^* gives $(iii) = \sqrt{T}\,\nabla(\hat{\Delta}_m - \Delta)(\theta^*)^\top(\bar{\theta}_T - \theta^*) + o_p(1)$; the two factors are $O_p(m^{-1/2})$ and $O_p(T^{-1/2})$ respectively (Lemma 1), so $(iii) = O_p(m^{-1/2}) = o_p(1)$ because $m \to \infty$ in the regime $T/m \to \lambda$. Term (ii) equals $\sqrt{T/m}\,\sqrt{m}(\hat{\Delta}_m(\theta^*) - \Delta(\theta^*))$, which by Lemma 2 converges to $\mathcal{N}(0, \lambda v^2)$ and is independent of (i) by assumption. Slutsky's lemma finishes the proof.

A.3 Proof of Corollary 1

Let $\sigma_{\mathrm{tot}}^2 = \sigma_\Delta^2 + \lambda v^2$. The interval of Roy and Mohapatra [2026] has half-width $z_{\alpha/2}\sigma_\Delta/\sqrt{T}$, hence

$$\Pr\{\hat{\Delta}_m(\bar{\theta}_T) \in I_T\} = \Pr\left\{\left|\frac{\sqrt{T}(\hat{\Delta}_m(\bar{\theta}_T) - \Delta(\theta^*))}{\sigma_{\mathrm{tot}}}\right| \leq z_{\alpha/2}\frac{\sigma_\Delta}{\sigma_{\mathrm{tot}}}\right\} + o(1) = 2\Phi\Big(z_{\alpha/2}\frac{\sigma_\Delta}{\sigma_{\mathrm{tot}}}\Big) - 1.$$

The ratio $\sigma_\Delta/\sigma_{\mathrm{tot}}$ equals one if and only if $\lambda v^2 = 0$, so the coverage equals $1 - \alpha$ if and only if $m \gg Tv^2/\sigma_\Delta^2$.

A.4 Proof of Remark 1

With a fixed evaluation set, the interval error is $\hat{\Delta}_m(\bar{\theta}_T) - \Delta(\theta^*) = (\Delta(\bar{\theta}_T) - \Delta(\theta^*)) + b$ with $b = \hat{\Delta}_m(\theta^*) - \Delta(\theta^*)$ fixed conditional on the evaluation set. By Lemma 1,

$$\Pr\left\{\left|\sqrt{T}b + \sqrt{T}(\Delta(\bar{\theta}_T) - \Delta(\theta^*))\right| \leq z_{\alpha/2}\sigma_\Delta\right\} = \Phi\big(z_{\alpha/2} - \sqrt{T}b/\sigma_\Delta\big) - \Phi\big(-z_{\alpha/2} - \sqrt{T}b/\sigma_\Delta\big) + o(1),$$

which tends to zero for $|b| > 0$; the simulations average this expression over the law of b, whose standard deviation is $v/\sqrt{m}$. This is the sense in which coverage statements about population fairness require a fresh evaluation set.

A.5 Proof of Theorem 2

Condition on the training data. For a training multiplier trajectory, $\sqrt{T}(\hat{\Delta}_m(\bar{\theta}_T^b) - \hat{\Delta}_m(\bar{\theta}_T)) = \sqrt{T}(\Delta(\bar{\theta}_T^b) - \Delta(\bar{\theta}_T)) + \sqrt{T}\big[(\hat{\Delta}_m - \Delta)(\bar{\theta}_T^b) - (\hat{\Delta}_m - \Delta)(\bar{\theta}_T)\big]$; by the online multiplier bootstrap theorem of Roy and Mohapatra [2026] the first term has conditional law converging to $\mathcal{N}(0, \sigma_\Delta^2)$, and the second is $o_p(1)$ by the same localisation argument as in the proof of Theorem 1, because both iterates lie within $O_p(T^{-1/2})$ of each other. The evaluation term $\sqrt{m}(\hat{\Delta}_{m,r}(\bar{\theta}_T) - \hat{\Delta}_m(\bar{\theta}_T))$ is a standard empirical bootstrap of an i.i.d. sample: conditional on the data, the multinomial reweighting gives a conditional law converging to $\mathcal{N}(0, v^2)$ by the same delta method as Lemma 2. The multiplier and evaluation resamplings use independent randomness, so the two conditional laws are asymptotically independent; after the rescaling of the main text this gives the law of $\sqrt{T}(\hat{\Delta}_m(\bar{\theta}_T) - \Delta(\theta^*))$. For the third component, each resampled constraint set yields θ_s^*; when the resampled sets are i.i.d. from the audit design, the empirical distribution over s of $\hat{\Delta}_m(\theta_s^*) - \hat{\Delta}_m(\theta^*)$ converges to its sampling law as $S \to \infty$, provided the constrained optima are estimated with $o_p(S^{-1/2})$ accuracy. The (modified) percentile interval is therefore asymptotically valid. The folded interval for $|\Delta(\theta^*)|$ has coverage at least that of the signed interval: if the signed interval $[l, u]$ contains $\Delta > 0$ then $u \geq \Delta$ and hence $\max(|l|, |u|) \geq |\Delta|$; the case $\Delta < 0$ is symmetric, and when the signed interval contains both signs the folded interval is $[0, \max(|l|, |u|)]$ and covers $|\Delta|$ trivially.

A.6 Derivation of the sample-size allocation

Minimising $c_T T + c_m m$ subject to $z_{\alpha/2}^2(\sigma_\Delta^2/T + v^2/m) \leq w^2$ has Lagrangian $c_T T + c_m m + \mu(\sigma_\Delta^2/T + v^2/m - w^2/z_{\alpha/2}^2)$. The first-order conditions give $T = \sigma_\Delta\sqrt{\mu/c_T}$ and $m = v\sqrt{\mu/c_m}$, hence $m/T = (v/\sigma_\Delta)\sqrt{c_T/c_m}$. Substituting into the constraint yields $\sqrt{\mu} = z_{\alpha/2}^2(\sigma_\Delta\sqrt{c_T} + v\sqrt{c_m})/w^2$ and therefore

$$T = \frac{z_{\alpha/2}^2}{w^2}\Big(\sigma_\Delta^2 + \sigma_\Delta v\sqrt{c_T/c_m}\Big), \qquad m = \frac{z_{\alpha/2}^2}{w^2}\Big(v^2 + \sigma_\Delta v\sqrt{c_T/c_m}\Big).$$

B Reproduction details

The reproduction follows the published equations, with the deviations listed in Table 7; the complete report, code and raw result files are released with this paper. No official implementation accompanies Roy and Mohapatra [2026], so the training procedures (weighted dual averaging with slab projection for DI, penalised SGD for DM) and the online multiplier bootstrap were re-implemented from the paper. The most consequential deviations are the step sizes: in the synthetic DI experiment the published step size $\eta_0 = 0.01$ leaves the averaged iterate far from the constrained optimum even at $T = 10^5$, so we used $\eta_0 = 0.1$; the synthetic DM generator scales the loss by $R^2 = 500$ and required $\eta_0 = 0.03$ to avoid divergence. On the real datasets we regularised by $\kappa = 0.1$ so that the constrained optimum is reached within the training horizon; this choice is analysed in Table 5. The complementary sensitivity analysis for the DM penalty effective at the averaged iterate is reported in the released report.

Table 7: Deviations of the reproduction from the published settings (all other settings as in the original Table 1). "z-score" refers to normalising real features before training.

setting	baseline	ours	reason
step size η_0 (synthetic DI)	0.01	0.1	convergence within $T = 2 \times 10^4$
step size η_0 (synthetic DM)	0.01	0.03	divergence under $R^2 = 500$ scaling
regularisation κ (real)	10^{-4}	0.1	reach constrained optimum
features (real)	raw	z-score	conditioning
repetitions (real)	100/200	100–150	compute budget
θ^*	trained	direct solve	audit-protocol baseline
population $\phi(\theta^*)$	held-out	full pool	estimand definition

References

Xi Chen, Jason D. Lee, Xin T. Tong, and Yichen Zhang. Statistical inference for model parameters in stochastic gradient descent. *The Annals of Statistics*, 48(5):2513–2540, 2020. doi: 10.1214/18-AOS1801.

John J. Cherian and Emmanuel J. Candès. Statistical inference for fairness auditing. *Journal of Machine Learning Research*, 25, 2024. arXiv:2305.03712; ACM DL 10.5555/3722577.3722726.

Sam Corbett-Davies, Emma Pierson, Avi Feller, Sharad Goel, and Aziz Huq. Algorithmic decision making and the cost of fairness. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 797–806, 2017. doi: 10.1145/3097983.3098095.

Cyrus DiCiccio, Sriram Vasudevan, Kinjal Basu, Krishnaram Kenthapadi, and Deepak Agarwal. Evaluating fairness using permutation tests. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1467–1477, 2020. doi: 10.1145/3394486.3403199.

Frances Ding, Moritz Hardt, John Miller, and Ludwig Schmidt. Retiring adult: New datasets for fair machine learning. *Advances in Neural Information Processing Systems*, 34:6478–6490, 2021. doi: 10.48550/arXiv.2108.04884.

John C. Duchi and Feng Ruan. Asymptotic optimality in stochastic optimization. *The Annals of Statistics*, 49(1):21–48, 2021. doi: 10.1214/19-AOS1831.

Yixin Fang, Jinfeng Xu, and Lei Yang. Online bootstrap confidence intervals for the stochastic gradient descent estimator. *Journal of Machine Learning Research*, 19(78):1–30, 2018.

Moritz Hardt, Eric Price, and Nathan Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016. arXiv:1610.02413.

Subha Maity, Songkai Xue, Mikhail Yurochkin, and Yuekai Sun. Statistical inference for individual fairness. In *International Conference on Learning Representations (ICLR)*, 2021. doi: 10.48550/arXiv.2103.16714.

Clara Moreau and Julia Weber. Quantifying algorithmic fairness: A novel perspective through uncertainty estimation. *European Journal of Emerging Artificial Intelligence*, 1(1):96–112, 2024. ISSN 3087-4602.

Boris T. Polyak and Anatoli B. Juditsky. Acceleration of stochastic approximation by averaging. *SIAM Journal on Control and Optimization*, 30(4):838–855, 1992. doi: 10.1137/0330046.

Pratik Ramprasad, Yuantong Li, Zhuoran Yang, Zhaoran Wang, Will Wei Sun, and Guang Cheng. Online bootstrap inference for policy evaluation in reinforcement learning. *Journal of the American Statistical Association*, 118(544):2901–2914, 2023. doi: 10.1080/01621459.2022.2096620.

Lucas Rosenblatt and R. Teal Witter. Fairlyuncertain: A comprehensive benchmark of uncertainty in algorithmic fairness. *arXiv preprint arXiv:2410.02005*, 2024. doi: 10.48550/arXiv.2410.02005.

Abhishek Roy and Prasant Mohapatra. Fairness uncertainty quantification: How certain are you that the model is fair? *arXiv preprint arXiv:2304.13950*, 2023. doi: 10.48550/arXiv.2304.13950.

Abhishek Roy and Prasant Mohapatra. Fairness uncertainty quantification: A constrained stochastic optimization perspective. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI)*, volume 337 of *PMLR*, pages 5845–5865, 2026. arXiv:2304.13950.

Bahar Taskesen, Jose Blanchet, Daniel Kuhn, and Viet Anh Nguyen. A statistical test for probabilistic fairness. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pages 648–665, 2021. doi: 10.1145/3442188.3445927.

Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez-Rodriguez, and Krishna P. Gummadi. Fairness constraints: A flexible approach for fair classification. *Journal of Machine Learning Research*, 20(75):1–42, 2019.

Wanrong Zhu, Xi Chen, and Wei Biao Wu. Online covariance matrix estimation in stochastic gradient descent. *Journal of the American Statistical Association*, 118(541):393–404, 2023. doi: 10.1080/01621459.2021.1933498.